

LEARNING OUTCOMES!

Candidates should be able to:

- describe the fishing methods used in both marine and inland waters, including fish farms
- give examples of the fish caught in both marine and inland waters, and of the fish reared on fish farms
- give examples of the fishing ports on both the Balochistan and Sindh coasts
- describe the uses of the fish caught
- explain improvements in fishing methods and processing techniques
- understand the problems facing the fishing industry and evaluate the possibilities for its further development and sustainability.

FISH FOOD

- ✓ Fish feed on planktons
- ✓ Planktons are minute plant and animal organisms that float in sea or are deposited on the sea bed
- ✓ Carbon dioxide and oxygen dissolves in surface water by sun rays from atmosphere
- ✓ Phosphates, nitrates and other plant nutrients are added to the surface water by rivers
- ✓ They together help the growth of planktons

ADVANTAGES OF FISHING INDUSTRY

- ✓ Source of nutritious food.
- ✓ Fish Oil
- ✓ Waste useful for fertilizer industry
- ✓ Increases GDP
- ✓ Export of fish can earn foreign exchange
- ✓ Source of income
- ✓ Source of direct and indirect employment
- ✓ Promotes the growth of other industries e.g. boat makers
- ✓ Development of coastal areas
- ✓ Waterlogged and saline areas can be used for fish farming

DISADVANTAGES OF FISHING INDUSTRY

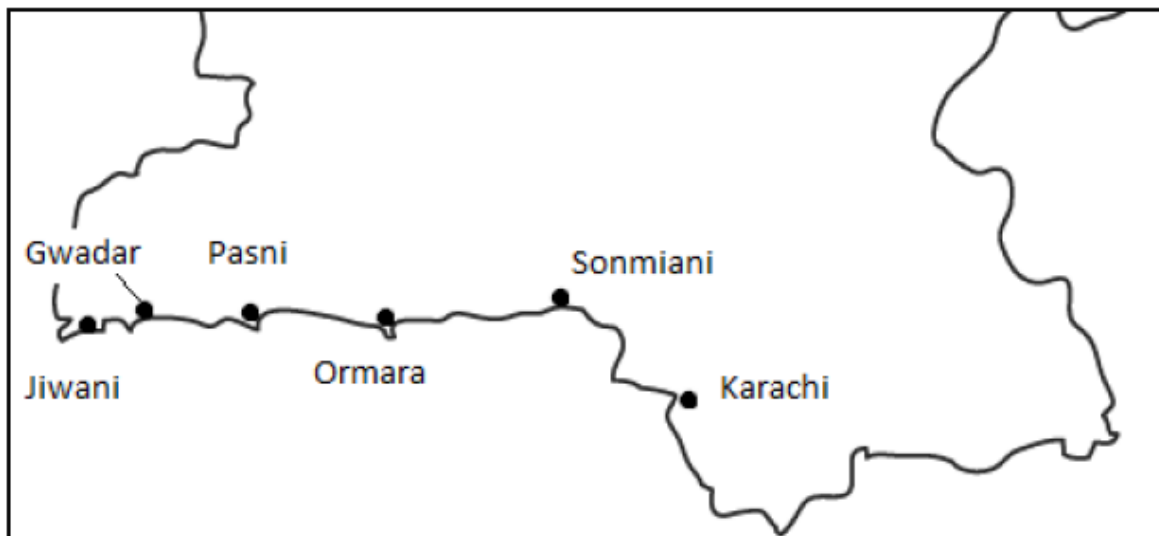
- ✓ Lack of investment so old methods followed.
- ✓ Poor infrastructure
- ✓ Lack of skilled labour
- ✓ Water pollution
- ✓ Issues of overfishing
- ✓ Less Profit

MARINE FISHING

- ✓ Carried out along with coastal water of Sindh and Baluchistan
- ✓ Sindh Coast is 30% of the coastline and Makran Coast is 70%

FISHING PORTS

- ✓ Sindh: Karachi, Thatta
- ✓ Baluchistan: Jiwani, Gwadar, Pasni, Ormara, Sonmiani



SUBSISTENCE FISHING

- ✓ Fishermen and his family consume the fish, they catch
- ✓ Small wooden boats under 20m long
- ✓ Traditional wooden boats
- ✓ Owner and its crewmen
- ✓ Upto 5 km off the shore
- ✓ Traditional nets, hand nets
- ✓ No Radar and Sonar to find fish
- ✓ One day operation
- ✓ No Satellites for navigation and weather
- ✓ Ice Box
- ✓ Traditional, weak ropes
- ✓ Gutted
- ✓ Dried
- ✓ Salted
- ✓ Nominal catch

COMMERCIAL FISHING

- ✓ Fish is sold in the market
- ✓ Boat size upto 100 m long
- ✓ Mechanized boats (With engine)
- ✓ As many as 100 people employed by a company
- ✓ Away from port upto 50-60 Km from the coast in the deep-sea water
- ✓ Nets at least 1 km wide. Gill netters are used.
- ✓ Radar and Sonar to find fish
- ✓ Away from port for weeks
- ✓ Satellites for navigation and weather
- ✓ Refrigerators
- ✓ Stronger bigger nylon nets and ropes
- ✓ Gutted
- ✓ Canned
- ✓ Dried
- ✓ Salted
- ✓ Fish catch is high

TYPES OF MARINE FISH CATCH

- ✓ Sharks
- ✓ Croakers
- ✓ Drums
- ✓ Cat fish
- ✓ Skates
- ✓ Rays

WHY SINDH COAST IS IMPORTANT FISHING CENTRE?

- ✓ Large market
- ✓ More interest from investors
- ✓ Good roads
- ✓ Reliable electricity
- ✓ Clean water
- ✓ Modern infrastructure
- ✓ Skilled labour
- ✓ Foreign investments
- ✓ Better port facilities
- ✓ Better cold storage
- ✓ More mechanized fishing fleet
- ✓ Numerous creeks
- ✓ Sheltered harbours
- ✓ Wide Shallow sea
- ✓ Wider continental shelf
- ✓ More area under mangroves
- ✓ Indus Delta is rich in fish food

FISH MARKET IN KARACHI

- ✓ Karachi had domestic and international market
- ✓ At domestic level fish is supplied to local markets through wholesale dealers
- ✓ Street hawkers also buy some of fish catch and provide a door-to-door service
- ✓ Frozen processed fish is supplied to large department stores
- ✓ About 30% of total fish catch is exported to 30 countries of the world
- ✓ Japan is main market for fish and shrimps
- ✓ UK, USA and France are other important market

FISH MARKETING IN GAWADAR

- ✓ Fish catch is packed in ice for transportation to Karachi
- ✓ From Karachi it is further exported
- ✓ Some businessmen from Karachi or foreign country buy fish in open sea to avoid paying port charges
- ✓ Fish is exported directly to middle East Countries
- ✓ Some of fish is marketed locally

How does poor infrastructure of Baluchistan make development of fishing difficult?

- ✓ Poor Roads/Railways for Transportation
- ✓ Lack of electricity for processing
- ✓ Poor Telecommunication so difficult to sell products
- ✓ Lack of clean water for processing
- ✓ Lack of education so difficult to follow new commercial methods

Advantages Of Developing Fish Processing Industries In Baluchistan

- ✓ Stimulates development of fishing industry
- ✓ Reduced cost of transport (than to Karachi)
- ✓ Reduces Burden on Karachi
- ✓ More fresh / no delay / no need for storage
- ✓ Infrastructure development, e.g. roads, power, telecommunications
- ✓ Sources of Income - higher living standards, better housing, jobs
- ✓ Trade with Middle East increases

Disadvantages Of Developing Fish Processing Industries In Baluchistan

- ✓ Lack of infrastructure, e.g. roads, power, water, ports, etc.
- ✓ Small market / population
- ✓ Long way from major centres of population, e.g. Karachi
- ✓ Uneducated / unskilled population
- ✓ Lack of interest from investors or government
- ✓ Inhospitable climate
- ✓ Poor quality product

FISHING PERIODS

- ✓ November to January for fish
- ✓ October and November for shrimps
- ✓ June and July is breeding time
- ✓ Fishing in big boats is done throughout the year
- ✓ Part time fishermen who use small boats do not fish from May to September because of South-West monsoon

IMPROVEMENTS IN MARINE FISHING INDUSTRY

- ✓ Training facilities are being provided
- ✓ Establishing on-shore facilities
- ✓ Better marketing and processing facilities
- ✓ Increasing the efficiency of existing boats
- ✓ Research surveys of deep off-shore waters are being conducted to promote deep sea fishing
- ✓ 16 ice factories are developed in Baluchistan
- ✓ Improved loading and unloading facilities
- ✓ Guidance and weather information for fishermen
- ✓ Monitoring of deep-sea fishing vessels

INLAND FISHING

- ✓ Practiced in almost all the rivers, lakes, reservoirs, irrigation channels and fish farms

TYPES OF INLAND FISH CATCH

- ✓ Manaseer
- ✓ Palla
- ✓ Thalla
- ✓ Rahu
- ✓ Trout
- ✓ Grass
- ✓ Silver

MAIN FISHING CENTERS

- ✓ Manchar Lake in Dadu District Sindh
- ✓ Keenjhar Lake North of Thatta
- ✓ Haleji Lake West of Thatta
- ✓ Reservoirs of Mangla and Tarbela Dam
- ✓ River Indus at Sukkur, Kotri, Thatta

FISH FARMING

- ✓ Flat land of plains easy to use machinery
- ✓ Large and cheap land is available to build the farms
- ✓ Unconsolidated rocks and alluvial deposits make easy to dig out ponds
- ✓ Higher water table which can use to fill the ponds
- ✓ Large irrigational schemes
- ✓ Govt. Incentives
- ✓ Many hatcheries which provide baby fish
- ✓ Good road networks for supply of fish
- ✓ Large domestic market
- ✓ Large rectangular man-made ponds
- ✓ Concrete or cemented impervious base and sides to prevent water loss
- ✓ Sides of farms and edged with solidified mud to prevent contamination
- ✓ Linear plantation around fish farms to provide oxygen, shade to minimize water loss by evaporation and to keep water body cool

- ✓ Separation of ponds by embankments for different ages and species of the fish
- ✓ Roads and ponds lined to prevent contamination



How Fishes are reared on a Fish Farm?

- ✓ Farmers bring baby fish from the hatcheries
- ✓ Regular feed for proper growth and healthier fish
- ✓ Regular checking
- ✓ Disinfectants to avoid diseases
- ✓ Regular check of water level
- ✓ Regular change of fresh water
- ✓ Different ages of fish are transferred between ponds
- ✓ When fish is mature and meet the need of market, they are caught by nets

Advantages

- ✓ Provides income
- ✓ Raw material for fertilizer industry
- ✓ Provided employment
- ✓ Reduces burden on crops and livestock
- ✓ Foreign exchange earnings by export

Disadvantages

- ✓ Occupies more space

- ✓ Lower water table
- ✓ Deforestation
- ✓ Skills needed
- ✓ Capital needed
- ✓ Continuous supply of fresh water
- ✓ Good network of roads

Why more people employed in inland fishing?

- ✓ More people live near ponds, lakes
- ✓ Maintenance of ponds
- ✓ Feeding
- ✓ Catching of Fish
- ✓ Hatcheries (Separate labour required)

PROBLEMS OF FISHING INDUSTRY

WATER POLLUTION

- ✓ Toxic chemicals can also accumulate in fish tissues, making fish too dangerous for human to eat
- ✓ Eutrophication: When phosphorous and nitrogen get into water causes growth of algae. Algae uses large amount of oxygen. Fish suffocate in oxygen depleted water
- ✓ Thermal Pollution: Cool water from river or lake is circulated around engines to cool it and then warm water is then returned to river or lake. It can cause massive fish kills when discharged water is too warm. Warm water hold less oxygen and fish may suffocate
- ✓ No proper dumping of domestic and industrial waste sea animals may eat plastic and die of suffocation or maybe they get strangled

OVERFISHING

- ✓ Overfishing means when more fish are caught than replaced naturally
- ✓ Fishing in breeding season
- ✓ Catching baby fishes
- ✓ Too many fishes caught

THREAT TO MANGROVES

- ✓ Mangroves act as barrier and protect coastline from high and low tides
- ✓ Mangroves are breeding ground for fish
- ✓ Mangroves cannot thrive in polluted water

GOVERNMENT MEASURES TO PROMOTE FISHERIES

- ✓ Establishment of fish hatcheries, nurseries and seed forms for large scale production of quality seed
- ✓ Improved management system
- ✓ Development of lakes, rivers
- ✓ Proper advisory services to private farmers
- ✓ Irrigation facilities and reduced rates
- ✓ Provision of bulldozers on reduced rates
- ✓ Supply of fish seed at nominal rates
- ✓ Free training
- ✓ Improvements of Karachi Harbour

- ✓ Construction of Korangi Fish Harbour, Pasni Fish Harbour
- ✓ Construction of Cold Storage
- ✓ Established shrimp packing and freezing plants in Karachi

HOW TO DEVELOP FISHERIES ON SUSTAINABLE LINES

- ✓ By avoiding overfishing
- ✓ By having quotas/government licences
- ✓ Secure fishing grounds from foreign fishing boats
- ✓ Limit number of months in year able to fish (to allow breeding)
- ✓ Only catch adult fish (by using nets with larger holes/mesh)
- ✓ Laws to protect of mangrove forests
- ✓ Laws/fines to prevent marine pollution
- ✓ Education in sustainable methods

TYPES OF NET FISHING

TRAWLING

- ✓ Makes use of a trawl net
- ✓ Both sides of trawl net are attached to one or more fishing vessels which drag it through deep water
- ✓ This method is used primarily to catch fish what live near bottom of sea

PURSE SEINING

- ✓ Uses a large net
- ✓ The boat surrounds school of fish with a curtain of netting and then close the bottom of net
- ✓ This is used to catch fish in the middle or at the surface of sea

DRIFT NETTING

- ✓ Makes use of a long net set vertically across the path of migrating fish
- ✓ When the fish try to make their way through net, they became entangled in it

PRIMARY SECTOR IN FISHING

- ✓ Fisherman

SECONDARY SECTOR IN FISHING

- ✓ Gutting
- ✓ Salting
- ✓ Canning
- ✓ Freezing
- ✓ Packaging

TERTIARY SECTOR IN FISHING

- ✓ Loading and unloading man
- ✓ Truck Driver

6 MARKS

Question 1**REVIEW BOOK BY MYM**

To what extent is it possible to develop the fish processing industry further in Pakistan? Give reasons to support your answer and refer to places or examples you have studied. [6]

Possible

- ✓ Long undeveloped coastline (1050km / Makran Coast 750 km)
- ✓ Gwadar being developed as a new port having fish harbour with modern facilities
- ✓ Potential at Pasni / Jiwani / Ormara (allowing more fish to be refrigerated / preserved for transport to Karachi)
- ✓ International quality standards (to remove import bans / embargoes)
- ✓ Increase local ice factories / refrigerated storage / packing / canning facilities (to reduce need to transport to Karachi)
- ✓ Training / education (could provide employment of local educated youth)
- ✓ Value added products made for export (make more foreign exchange)

Not possible

- ✓ Limited private sector and government investment
- ✓ Many processing plants under capacity or out of operation (showing that the future development is uncertain)
- ✓ Few skilled workers
- ✓ Coastline areas are remote and poor transport links
- ✓ Canning factories have been unhygienic and a cause for import bans (to EU / Saudi Arabia)
- ✓ Unreliable export market (about 30% worldwide)
- ✓ Low profits (6% of foreign exchange)
- ✓ Foreign competition
- ✓ Urban centres prefer fresh fish so processed fish only to a few large department stores
- ✓ Per capita consumption is low (1.6 kg p.a.)

Question 2**REVIEW BOOK BY MYM**

To what extent is it possible for marine fishing to be developed sustainably in Pakistan? Give reasons to support your answer. [6]

Possible

- ✓ By avoiding overfishing
- ✓ By having quotas/government licences
- ✓ Secure fishing grounds from foreign fishing boats
- ✓ Limit number of months in year able to fish (to allow breeding)
- ✓ Only catch adult fish (by using nets with larger holes/mesh)
- ✓ Laws to protect of mangrove forests
- ✓ Laws/fines to prevent marine pollution
- ✓ Education in sustainable methods

Not possible

- ✓ Mangrove forests are being cleared (reducing breeding/feeding areas for fish/shrimps)
- ✓ Sea pollution (oil from ships/industrial/domestic waste from Karachi)(poisoning fish and spreading into food chain)
- ✓ Not enough capital/investment
- ✓ Lack of skills/training in sustainable methods
- ✓ Lack of political interest/will (fishing only makes up about 0.5% of GDP)

Question 3**REVIEW BOOK BY MYM**

Fishing is an important and growing economic activity in Pakistan. Read the following two views about ways to develop the fishing industry in Pakistan sustainably:

A

Developing inland fish farming has greater benefits for the people and economy and is more sustainable.

B

Developing marine fishing has greater benefits for the people and economy and is more sustainable.

I agree with view B more.

Why?

- ✓ Marine fishing do not need any kind of investment, however inland fishing requires investment and Pak do not have funds.
- ✓ Improved facilities are being developed along the coast for preservation and storage by the government however for inland we have to do everything on own.
- ✓ Infrastructure facilities provided at Gwadar, Ormara and Pasni
- ✓ Ice factories and cold storage developed at Gwadar.
- ✓ One fish harbour has been constructed at Pasni.